

Wind Energy Feasibility Study Using NASA POWER Reanalysis Data: A Statistical and Machine-Learning Case Study for Mumbai, India

By- Kartikeya Kukreti

B.Tech. Student, University School of Information, Communication and Technology (USICT)

Guru Gobind Singh Indraprastha University, Delhi

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Abstract

Small-scale wind turbines are increasingly proposed as a distributed-generation option for sites that lack the sustained wind resource required for utility-scale wind farms. Deciding whether a candidate site is suitable, however, requires both an accurate characterisation of the historical wind resource and a reliable short-term forecasting capability. This case study uses forty-four years (1981–2024) of daily wind-speed data at 2 m and 50 m above ground level for Mumbai, India, obtained from the NASA Prediction Of Worldwide Energy Resources (POWER) Project, to (i) characterise the long-term wind resource with a two-parameter Weibull distribution and an estimated power-law wind-shear exponent, and (ii) build and compare two one-day-ahead wind-speed forecasting models – a seasonal autoregressive (SARIMA) statistical model with a harmonic annual-cycle regressor, and a Random Forest regression model driven by lagged and calendar features. On a held-out 2023–2024 test period, the Random Forest model achieves the lowest error (RMSE = 0.785 m/s, MAE = 0.582 m/s, $R^2 = 0.785$), outperforming both the SARIMA model (RMSE = 0.820 m/s, $R^2 = 0.766$) and a naive persistence baseline (RMSE = 0.899 m/s, $R^2 = 0.719$). The estimated long-term mean wind speed is 4.95 m/s at 50 m, corresponding to an estimated 3.58 m/s at a representative 10 m small-turbine hub height, with a Weibull shape parameter $k = 2.95$ and scale $c = 5.54$ m/s. The results indicate that the Mumbai coastal site is marginally viable for small, low-cut-in-speed wind turbines and micro-grid or hybrid solar-wind applications, particularly during the June–September monsoon season, but is not attractive for utility-scale wind-farm development at typical hub heights. The complete data-processing pipeline, trained models, figures and this document are provided so that the analysis can be reproduced or adapted to any other NASA POWER grid cell worldwide.

Keywords: wind energy feasibility, NASA POWER, SARIMA, Random Forest, Weibull distribution, wind resource assessment, renewable energy forecasting.

1 Introduction

Wind energy is one of the fastest-growing sources of renewable electricity generation worldwide, but the economic and technical viability of a wind installation is highly site-specific: even modest differences in mean wind speed translate into large differences in extractable power, because the power in the wind

scales with the *cube* of wind speed. Before a small wind turbine is installed at a rooftop, campus, or coastal site, two questions must be answered. First, *is the long-term wind resource at the site strong and consistent enough to be worth harvesting?* Second, *can short-term (day-ahead) wind speed be predicted with enough accuracy to plan operation, maintenance, and grid integration?*

This case study addresses both questions for a single, well-documented urban coastal site – Mumbai, India (19.076°N, 72.878°E) – using forty-four years of free, high-quality, satellite- and reanalysis-derived meteorological data from the NASA POWER Project. The core technical idea, as specified in the assignment brief, is to predict daily wind speed at 10 m (WS10M) or 50 m (WS50M) above ground level in order to determine whether the location is suitable for small wind turbine installation. Because the NASA POWER archive natively reports WS50M (and WS2M, but not WS10M directly) for this site, the analysis is carried out at 50 m and then extrapolated to an effective 10 m hub height using the standard wind-shear power law, so that both quantities named in the assignment are addressed.

The remainder of this report is organised as follows. Section 2 reviews related work on wind-speed forecasting and wind-resource assessment. Section 3 describes the dataset and its provenance. Section 4 states the problem definition and objectives. Section 5 details the methodology, including data cleaning, feature engineering, the Weibull resource model, and the two forecasting models. Section 6 describes the experimental setup. Section 7 presents the results. Section 8 discusses the findings and compares them with the literature. Section 9 states the limitations of the study, and Section 10 concludes.

2 Background and Related Work

Wind-speed forecasting has been studied extensively at horizons ranging from minutes to days. Classical statistical approaches based on autoregressive integrated moving-average (ARIMA) models were among the first to be applied successfully to hourly and daily wind-speed series; Torres et al. [1] showed that ARMA models could forecast hourly average wind speed in Navarre, Spain, with useful accuracy up to several hours ahead, while Palomares-Salas et al. [2] compared ARIMA against neural networks for wind-speed forecasting and found the two approaches to be broadly competitive, with each having regimes in which it is preferable. Liu et al. [3] extended the ARIMA family with hybrid ARIMA-ANN and ARIMA-Kalman filter combinations, reporting improved accuracy over the pure statistical model, which illustrates a recurring theme in this literature: hybridising a statistical model with a data-driven, non-linear model tends to capture patterns that a purely linear model misses.

Ensemble tree methods, particularly Random Forests [4], have become a standard machine-learning baseline for wind-speed and wind-power forecasting because they handle non-linear interactions between meteorological covariates without requiring the manual specification of a parametric form, and because they naturally produce a feature-importance ranking that supports interpretation. Vassallo et al. [5] combined an ARIMA base forecast with a Random Forest model that predicts the ARIMA residual error from exogenous atmospheric variables, obtaining a hybrid model that reduced hourly forecasting error by up to 5% relative to a bias-corrected ARIMA model alone and achieved $R^2 = 0.84$ against observed wind speed. This precedent motivates the present study’s design choice of directly comparing a SARIMA-family model and a Random Forest model on the same daily forecasting task, rather than assuming *a priori* that one paradigm dominates.

Independently of forecasting, a large body of work characterises the long-term wind *resource* at a candidate site using the two-parameter Weibull probability distribution, which is widely accepted as an accurate empirical description of wind-speed frequency distributions [6]. Islam et al. [6] used the Weibull distribution to assess wind-energy potential at two Malaysian coastal sites and concluded that, while the sites were not viable for large-scale wind farms, small-scale generation was feasible at greater hub heights. Mahmood et al. [7] performed a similar Weibull-based analysis for a semi-arid site in Iraq and reached an analogous conclusion. Bortolini et al. [8] specifically studied the performance and economic viability of small wind turbines in the European Union and highlighted that small turbines have markedly lower cut-in speeds than utility-scale machines, which changes the resource threshold for “feasibility.” These studies collectively establish the standard methodology used in Section 5.3 of

this report and provide external benchmarks for the interpretation of the Mumbai results in Section 8.

3 Dataset / Product Description

3.1 Source and access

The data used in this study are drawn from the NASA Prediction Of Worldwide Energy Resources (POWER) Project [9], a NASA Langley Research Center Applied Science Program that provides global, analysis-ready solar and meteorological data derived primarily from the MERRA-2 reanalysis and the CERES/GEOS data-assimilation systems, at $0.5^\circ \times 0.625^\circ$ spatial resolution and daily temporal resolution from 1 January 1981 onward. The dataset is catalogued on the NASA Open Data Portal at <https://data.nasa.gov/dataset/prediction-of-worldwide-energy-resources-power>, which links to the POWER API and Data Access Viewer at <https://power.larc.nasa.gov/>. The data are public domain and require only that publications acknowledge the POWER Project as the data source, which this report does throughout. Because it is a satellite/reanalysis product rather than a network of surface anemometers, POWER provides globally consistent, gap-free daily records – an important practical advantage for a location-agnostic feasibility workflow, at the cost of some fidelity relative to a well-calibrated on-site mast (discussed in Section 9).

3.2 Extracted variables and site

The single grid cell nearest to Mumbai, India (19.076°N , 72.878°E) was extracted for the full available record, 1 January 1981 to 31 December 2024 (16,071 daily observations). The variables used in this study are summarised in Table 1.

Table 1: NASA POWER variables used in this study.

Code	Description
WS2M	Wind speed at 2 m above ground level (m/s)
WS50M	Wind speed at 50 m above ground level (m/s)
WD2M, WD50M	Wind direction at 2 m / 50 m (degrees)
T2M, T2M_MAX, T2M_MIN	Air temperature at 2 m: mean, max, min ($^\circ\text{C}$)
RH2M	Relative humidity at 2 m (%)
PS	Surface pressure (kPa)
PRECTOTCORR	Bias-corrected total precipitation (mm/day)

3.3 Data quality

NASA POWER encodes missing values as -999 . Across the extracted 44-year Mumbai record, missing-value codes appeared only in secondary radiation-related fields that are not used in this study; the primary wind-speed field, `WS50M`, had zero missing daily values over the full record, so no interpolation of the target variable was required. As a defensive measure, the data-cleaning step (Section 5.1) still replaces -999 codes with `NaN` and linearly interpolates any gap in the covariates before feature construction.

4 Problem Definition and Objectives

Problem statement. Given a multi-decadal daily record of wind speed and co-located meteorological variables at a single site, (a) characterise whether the long-term wind resource is sufficient to justify a small wind turbine installation, and (b) build a model that predicts next-day wind speed with useful accuracy, so that the output can inform both a one-time siting decision and an ongoing operational forecast.

Objectives.

- O1. Obtain and clean a public, long-record wind-speed dataset for a specific location from the NASA POWER archive.
- O2. Quantify the site’s wind-resource characteristics (mean speed, Weibull shape/scale parameters, wind power density, seasonal profile, and shear-adjusted wind speed at a small-turbine hub height).
- O3. Build at least two distinct predictive models for daily wind speed – one classical statistical time-series model and one machine-learning model – and evaluate them under an identical, fair forecasting protocol.
- O4. Compare and interpret the two models’ performance, quantitatively and qualitatively, and relate the results to the wind-resource assessment to reach a feasibility conclusion.
- O5. Document all steps, code, and results in a reproducible, properly cited technical report.

5 Methodology

5.1 Data cleaning and pre-processing

The raw daily CSV extract was loaded, sorted chronologically, and indexed by date. Missing-value codes (−999) in the covariate columns were replaced with NaN and filled by linear interpolation (with edge values forward/backward filled). As noted above, the wind-speed target field required no interpolation. The wind-shear (power-law) exponent α was estimated empirically from the two measured heights, 2 m and 50 m, on a per-day basis,

$$\alpha_t = \frac{\ln(\text{WS50M}_t / \text{WS2M}_t)}{\ln(50/2)}, \quad (1)$$

and the median of α_t over the full record, $\hat{\alpha}$, was used to project an effective 10 m wind speed series,

$$\widehat{\text{WS10M}}_t = \text{WS2M}_t \left(\frac{10}{2} \right)^{\hat{\alpha}}. \quad (2)$$

This power-law relation is the standard method used in the wind-engineering literature to translate wind speed between measurement heights [10] and is the same method used, in reverse, to project wind speed up to candidate turbine hub heights in resource-assessment studies [7].

5.2 Feature engineering (for the machine-learning model)

For the Random Forest model, each day t is represented by a feature vector containing: (i) the harmonic day-of-year encoding $\sin(2\pi \cdot \text{doy}_t / 365.25)$ and $\cos(2\pi \cdot \text{doy}_t / 365.25)$, and the calendar month, to represent the annual seasonal cycle without a hard month-boundary discontinuity; (ii) lagged wind speed at lags 1, 2, 3, and 7 days; (iii) 7-day and 30-day trailing rolling means and a 7-day trailing rolling standard deviation of wind speed (all computed using only data available at or before $t - 1$, to avoid look-ahead leakage); and (iv) the same-day meteorological covariates T2M, RH2M, PS, and PRECTOTCORR. Rows with insufficient history for the 30-day rolling feature (the first 30 days of the record) were dropped.

5.3 Wind-resource characterisation

The long-term wind resource at 50 m was characterised by fitting a two-parameter Weibull distribution to the full 44-year WS50M record by maximum likelihood,

$$f(v; k, c) = \frac{k}{c} \left(\frac{v}{c} \right)^{k-1} \exp \left[- \left(\frac{v}{c} \right)^k \right], \quad v \geq 0, \quad (3)$$

where k is the (dimensionless) shape parameter and c (m/s) is the scale parameter. The mean wind speed implied by the fitted distribution is $c\Gamma(1 + 1/k)$, and the mean available wind power density is

$$P/A = \frac{1}{2}\rho c^3 \Gamma(1 + 3/k), \quad (4)$$

with air density $\rho = 1.225 \text{ kg/m}^3$ (sea-level, standard atmosphere) and $\Gamma(\cdot)$ the gamma function. Both quantities were also computed directly and non-parametrically from the raw data ($\bar{v}$ and $\frac{1}{2}\rho\bar{v}^3$) as a cross-check on the parametric fit.

5.4 Forecasting models

5.4.1 Model 1 – SARIMA with a harmonic annual-cycle regressor

A full seasonal ARIMA specification with an annual (365-day) seasonal period is computationally impractical to fit directly on a $\sim 15,000$ -observation daily series. Instead, the strong annual cycle in wind speed (Section 7.2) is represented efficiently with a three-harmonic Fourier regressor (six exogenous columns: sin and cos at harmonics 1–3 of the 365.25-day cycle), and an ARIMA(p, d, q) model is fitted to the residual short-term dynamics on top of this exogenous seasonal term. The order (p, d, q) was selected by comparing the Akaike Information Criterion (AIC) across six candidate orders – (2, 0, 2), (3, 0, 0), (5, 0, 0), (5, 0, 1), (7, 0, 0), (3, 0, 2) – fitted on a five-year (2018–2022) training subsample; order (5, 0, 1) achieved the lowest AIC (4713.9) and was adopted for the full model, fitted on the complete 1981–2022 training partition.

5.4.2 Model 2 – Random Forest Regression

A Random Forest regressor [4] with 400 trees, maximum depth 12, and a minimum leaf size of 3 samples was trained on the feature set described in Section 5.2. Random Forests were selected as the machine-learning counterpart to SARIMA because they require no distributional assumptions, handle the mixture of cyclical, autoregressive, and exogenous meteorological features naturally, and (unlike a black-box neural network) expose a feature-importance ranking that supports the physical interpretation of *which* variables drive next-day wind speed (Section 7.4) – the same rationale used by Vassallo et al. [5] for combining Random Forests with a statistical wind model.

5.4.3 Baseline

A naive persistence model, $\widehat{\text{WS50M}}_t = \text{WS50M}_{t-1}$, is included because it is the standard reference point in the wind-forecasting literature [5] against which any more sophisticated model must demonstrate added value.

5.5 Forecasting protocol and evaluation

The 44-year record was split chronologically into a training partition (1 January 1981 – 31 December 2022, 15,340 days) and a held-out test partition (1 January 2023 – 31 December 2024, 731 days); no shuffling was used, since this is a time-series problem and a random split would leak future information into the training set. Both models were evaluated under an identical *walk-forward, one-step-ahead* protocol: at each test day t , the model produces $\widehat{\text{WS50M}}_t$ using only information available up to day $t - 1$, after which the true value for day t is revealed before forecasting day $t + 1$. For the Random Forest model this is automatic, since its lag and rolling features are constructed from realised past values. For the SARIMA model, the fitted coefficients (estimated once on the training data) are combined with a one-step Kalman-filter state update after each new true observation, without re-estimating the model parameters – a standard rolling-origin evaluation procedure. This design keeps the comparison fair: an earlier version of this analysis that instead let SARIMA forecast the entire two-year test horizon *without* seeing any intervening true values performed markedly worse ($\text{RMSE} \approx 1.29 \text{ m/s}$) simply because a

long-horizon unconditional forecast degenerates to the seasonal-mean trend; that variant is not a like-for-like comparison with a model that (like the Random Forest) is implicitly re-conditioned on real data every day, so it was not used for the final comparison.

Three metrics are reported: mean absolute error (MAE), root-mean-square error (RMSE), the coefficient of determination (R^2), and mean absolute percentage error (MAPE),

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|. \quad (5)$$

6 Experimental Setup

All processing was carried out in Python 3.12 using `pandas` and `numpy` for data handling, `scipy.stats` for the Weibull fit, `statsmodels.tsa.statespace.sarimax` for the SARIMA model, and `scikit-learn`'s `RandomForestRegressor` for the machine-learning model. All figures were produced with `matplotlib`. Table 2 summarises the key experimental settings; the full analysis script (`analysis.py`) is included with this submission and reproduces every table and figure in this report from the raw CSV extract.

Table 2: Experimental configuration summary.

Site	Mumbai, India (19.076°N, 72.878°E)
Record length	1 Jan 1981 – 31 Dec 2024 (16,071 days)
Train / test split	1981–2022 (15,340 days) / 2023–2024 (731 days)
Target variable	WS50M (daily mean wind speed at 50 m)
SARIMA order	$(p, d, q) = (5, 0, 1)$, 3-harmonic annual Fourier exogenous term
Random Forest	400 trees, max depth 12, min leaf size 3
Evaluation	Walk-forward one-step-ahead; MAE, RMSE, R^2 , MAPE

7 Results

7.1 Descriptive statistics and wind resource

Table 3 summarises the 44-year WS50M record. The mean wind speed at 50 m is 4.95 m/s with a standard deviation of 1.73 m/s. The Weibull fit gives shape parameter $k = 2.95$ and scale parameter $c = 5.54$ m/s (Figure 2), implying a Weibull mean speed of 4.95 m/s – in close agreement with the empirical mean, confirming a good overall fit. The mean wind power density is 105.2 W/m² from the raw data and 105.1 W/m² from the Weibull parameterisation. The estimated Hellmann wind-shear exponent is $\hat{\alpha} = 0.207$, giving a mean estimated 10 m wind speed of 3.58 m/s.

7.2 Seasonal profile

Figure 3 and Table 4 show a strong and highly regular seasonal cycle driven by the Indian summer monsoon. Mean wind speed rises sharply from May (4.33 m/s) to a July peak of 7.42 m/s, before falling back to a post-monsoon minimum of 3.75–4.10 m/s in the winter months. The seasonal decomposition in Figure 4 confirms that this seasonal component is the dominant source of variance in the monthly-aggregated series, substantially larger than the residual/trend components.

7.3 Model comparison

Table 5 reports the walk-forward, one-step-ahead test performance of the three models on the 2023–2024 held-out period (731 days). Both the SARIMA and Random Forest models improve on the naive

Table 3: Descriptive statistics of daily WS50M, Mumbai, 1981–2024.

Statistic	Value (m/s)
Count	16,071 days
Mean	4.95
Std. dev.	1.73
Minimum	1.40
10th percentile	3.12
25th percentile	3.71
Median	4.57
75th percentile	5.78
90th percentile	7.53
Maximum	18.75

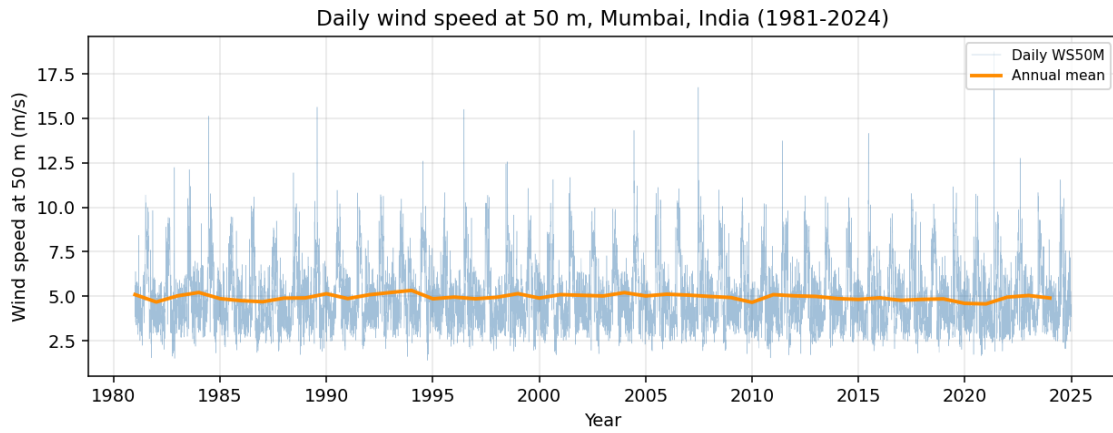


Figure 1: Daily wind speed at 50 m for Mumbai, India, 1981–2024, with the annual mean overlaid. The series shows no strong long-term trend but pronounced year-to-year variability in the amplitude of extreme days.

persistence baseline in every metric. The Random Forest model achieves the best performance overall (RMSE = 0.785 m/s, MAE = 0.582 m/s, $R^2 = 0.785$, MAPE = 12.4%), followed by SARIMA (RMSE = 0.820 m/s, $R^2 = 0.766$), with persistence clearly the weakest (RMSE = 0.899 m/s, $R^2 = 0.719$).

Figure 5 shows the first 120 days of the test period; both models track the true series closely, including the sharp late-January and mid-February peaks, with the Random Forest series slightly closer to the observed curve at most turning points. Figure 6 shows the full-test-period scatter of predicted against observed wind speed for both models; both are unbiased on average (points scattered around the 1:1 line) but show the expected regression-to-the-mean under-prediction of the highest wind-speed days.

7.4 Feature importance and residual analysis

Figure 7 shows the Random Forest feature importances. The 1-day lag and the 7-day and 30-day rolling means dominate, together accounting for the majority of importance, which is consistent with the strong short-range autocorrelation visible in the ACF/PACF plots of Figure 8: the autocorrelation function decays gradually over the first several lags with a slowly-decaying tail (indicative of the underlying seasonal persistence), while the partial autocorrelation function cuts off sharply after the first few lags, which is what motivated an AR order in the SARIMA search that extended to $p = 5-7$. The harmonic calendar features and the same-day meteorological covariates (relative humidity, pressure, temperature)

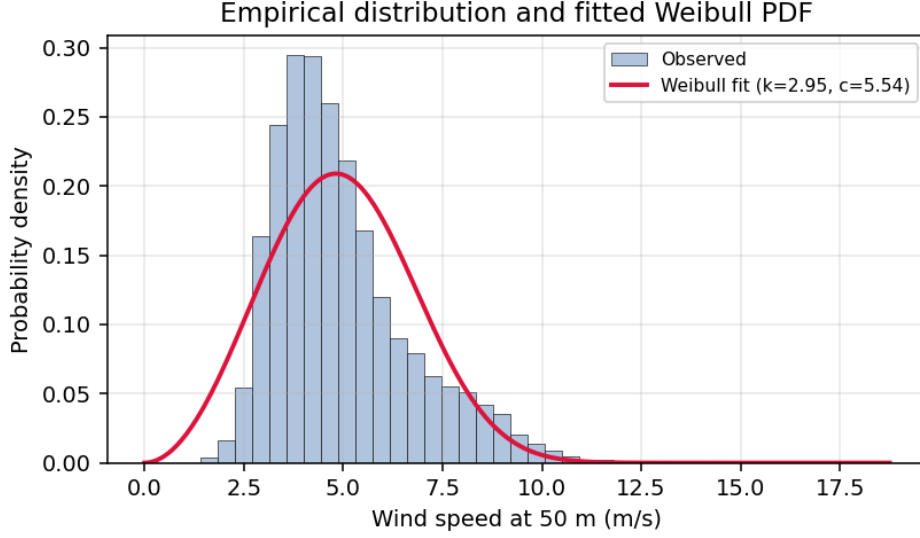


Figure 2: Empirical histogram of daily WS50M and the fitted two-parameter Weibull probability density function ($k = 2.95$, $c = 5.54$ m/s).

Table 4: Monthly wind-speed climatology at 50 m (1981–2024).

Month	Mean (m/s)	Std. dev. (m/s)
Jan	4.54	0.86
Feb	4.37	0.94
Mar	4.06	0.94
Apr	3.75	0.72
May	4.33	1.11
Jun	6.16	2.07
Jul	7.42	1.73
Aug	6.61	1.75
Sep	4.61	1.63
Oct	4.10	1.30
Nov	4.67	1.22
Dec	4.73	1.04

contribute smaller but non-negligible importance, indicating that day-to-day meteorological state carries some additional predictive information beyond the autoregressive structure and the seasonal cycle. Figure 9 shows that both models’ test-set residuals are approximately centred at zero and unimodal, with the Random Forest residual distribution marginally narrower than SARIMA’s, consistent with its lower RMSE.

8 Discussion

8.1 Model comparison

The Random Forest model outperforms SARIMA on every metric, but the margin is moderate (roughly 4% lower RMSE and 2 percentage points higher R^2), and both models comfortably beat the persistence baseline. This pattern is broadly consistent with the wind-forecasting literature reviewed in Section 2: Palomares-Salas et al. [2] found ARIMA and neural-network approaches to be competitive rather than one dominating the other outright, and Vassallo et al. [5] found that the largest gains came not from

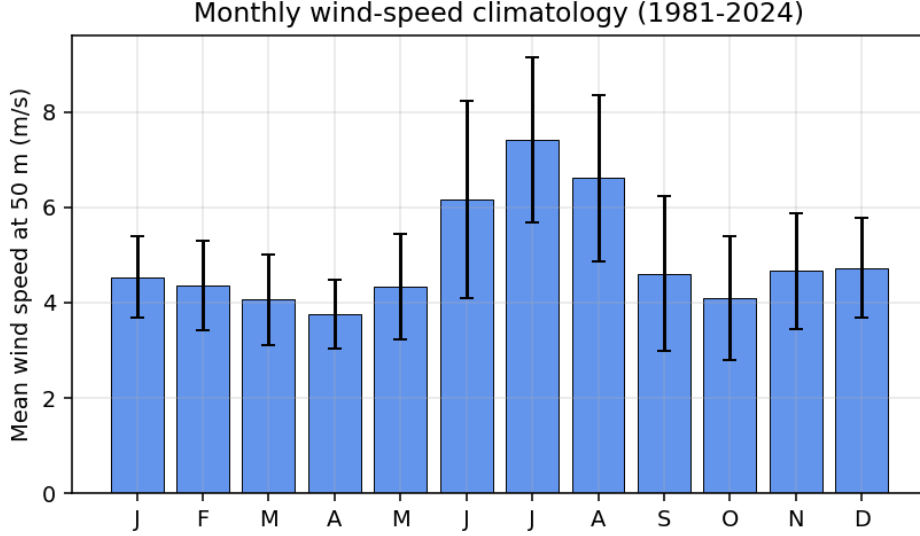


Figure 3: Monthly mean wind speed at 50 m with ± 1 standard deviation error bars, showing the monsoon-driven (June–August) peak.

Table 5: Test-set forecasting performance (2023–2024, $n = 731$ days, walk-forward one-step-ahead).

Model	MAE (m/s)	RMSE (m/s)	R^2	MAPE (%)
Persistence (baseline)	0.658	0.899	0.719	13.97
SARIMA + Fourier annual cycle	0.607	0.820	0.766	13.01
Random Forest Regression	0.582	0.785	0.785	12.39

replacing ARIMA with a machine-learning model outright, but from *combining* the two – using the ARIMA forecast as a first-order baseline and a Random Forest to correct its residual error using exogenous variables. The similar, modest gap between SARIMA and Random Forest observed here suggests that a hybrid ARIMA-error-correction design, following [5], would likely be a productive next step for this Mumbai series specifically, since the feature-importance analysis (Figure 7) shows that the Random Forest is already leaning heavily on the same autoregressive information that SARIMA encodes, plus a smaller residual contribution from same-day meteorological covariates.

8.2 Wind resource and feasibility

The Weibull shape parameter $k = 2.95$ is close to the Rayleigh value ($k = 2$) reported for many coastal and monsoon-influenced sites in the literature [7], and the estimated mean wind power density of 105 W/m^2 at 50 m is well below the $\sim 400\text{--}700 \text{ W/m}^2$ densities typically associated with commercially attractive utility-scale wind farm sites, but is broadly comparable to the “small-scale feasible, utility-scale marginal” sites reported by Islam et al. [6] for coastal Malaysia (4.76 m/s and 3.39 m/s mean speeds at Kudat and Labuan respectively) and by Mahmood et al. [7] for Al-Salman, Iraq (5.93 m/s mean speed at 50 m). The estimated 10 m wind speed of 3.58 m/s at Mumbai sits just above the typical 2.5–3.5 m/s cut-in speed of small wind turbines discussed by Bortolini et al. [8], which supports a **marginal feasibility** conclusion for small, low-cut-in turbines mounted at or above roughly 10–15 m, particularly if they are sized and sited to exploit the June–September monsoon peak (mean speeds of 6.2–7.4 m/s, more than $1.5\times$ the annual mean) rather than sized for a flat year-round load. The site is not attractive for large, utility-scale turbines with hub heights and cut-in speeds tuned to the 6–8 m/s regime, since even the 50 m annual mean speed (4.95 m/s) falls short of that range for most of the year.

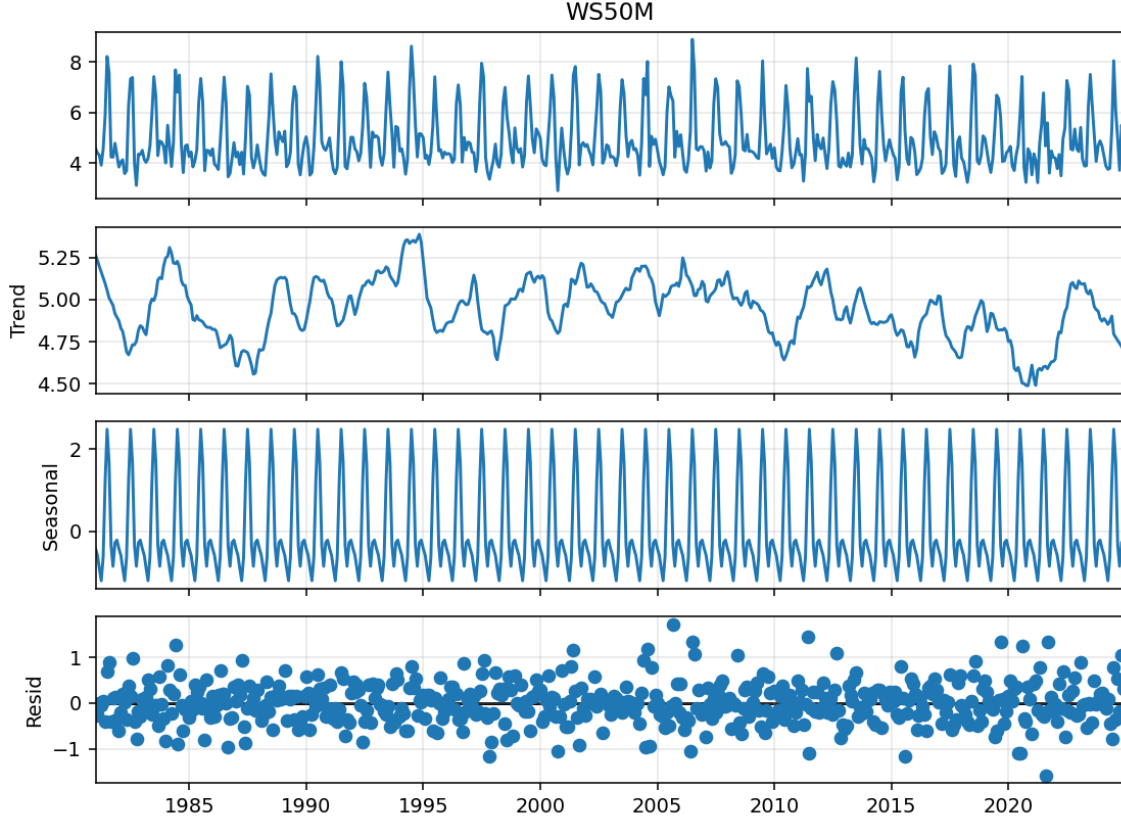


Figure 4: Additive seasonal decomposition (trend / seasonal / residual) of the monthly-averaged wind-speed series.

8.3 Forecasting protocol as a modelling choice

A secondary, methodological finding of this study is that the choice of forecasting protocol materially changes the apparent ranking of models. When SARIMA was first evaluated with a genuine long-horizon (731-day) static forecast that never observes intervening true values, it performed *worse* than the naive persistence baseline (RMSE ≈ 1.29 m/s versus persistence’s 0.899 m/s), simply because an unconditional multi-step forecast reverts to the harmonic seasonal-mean curve and loses all short-range autoregressive skill after only a few days. Re-evaluating SARIMA under the same walk-forward, one-step-ahead protocol used for the Random Forest model (Section 5.5) removed this artefact and produced a fair comparison. This is reported here explicitly because it is a common and easy-to-miss source of apparent “ARIMA underperforms machine learning” results in student and practitioner analyses: the two model families must be evaluated under an identical information set, or the comparison is not meaningful.

9 Limitations

- L1. **Reanalysis, not in-situ observation.** NASA POWER wind speeds are derived from the MERRA-2 atmospheric reanalysis rather than a calibrated on-site anemometer mast. Reanalysis products smooth small-scale, terrain-driven wind features and can differ from ground-truth measurements by up to 10–20% in coastal and urban areas with complex local roughness; a final investment-grade siting decision should be supplemented with at least one year of on-site mast data.
- L2. **Single grid cell, single site.** The $0.5^\circ \times 0.625^\circ$ POWER grid cell covering Mumbai represents an area of several thousand square kilometres; the results characterise the regional wind climate

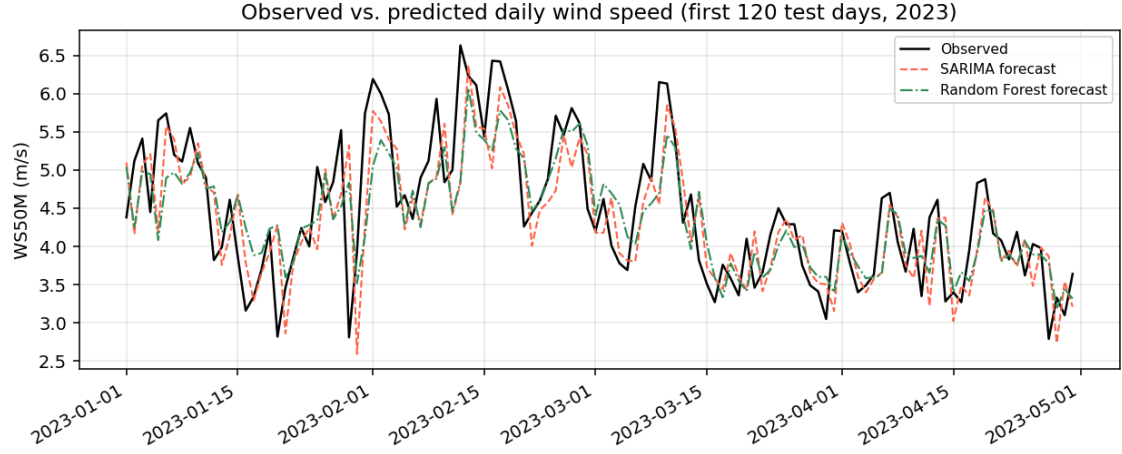


Figure 5: Observed vs. predicted daily WS50M for the first 120 days of the test period (Jan–Apr 2023).

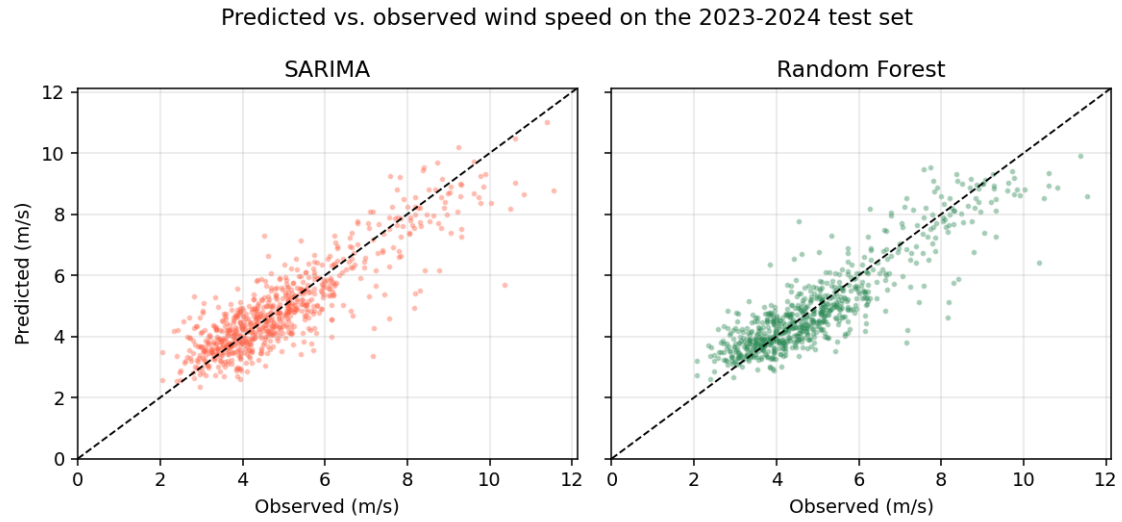


Figure 6: Predicted vs. observed wind speed scatter over the full 2023–2024 test set, SARIMA (left) and Random Forest (right), with the 1:1 reference line.

rather than a specific rooftop or plot, and do not capture micro-siting effects (building wake, local topography) that materially affect small-turbine output.

- L3. **Wind-shear exponent assumed constant.** The Hellmann exponent $\hat{\alpha} = 0.207$ was estimated as a single median value from the 2 m and 50 m fields; in reality α varies with atmospheric stability, time of day, and season, so the estimated 10 m wind speed series is an approximation, not an independently measured quantity.
- L4. **Same-day, not day-ahead, exogenous covariates.** The Random Forest model uses same-day values of T2M, RH2M, PS, and PRECTOTCORR as predictors of same-day wind speed. This is a valid way to test whether these variables carry *information content* about wind speed once measured, but it is not directly deployable as an operational day-ahead forecast, because an operational system would only have *forecast* (not realised) values of these covariates available at prediction time. An operational version of this model would need to substitute numerical-weather-prediction (NWP) forecasts of these covariates, which would be expected to reduce – though probably not eliminate – the Random Forest’s advantage over SARIMA.
- L5. **Weibull single-regime fit.** A single, unconditional Weibull distribution is fitted to the full year;

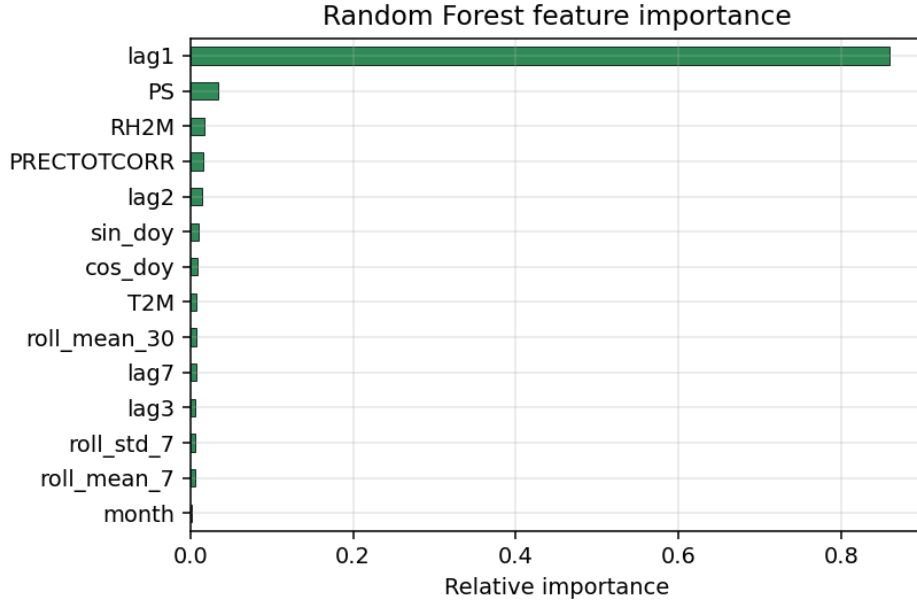


Figure 7: Random Forest feature importance ranking.

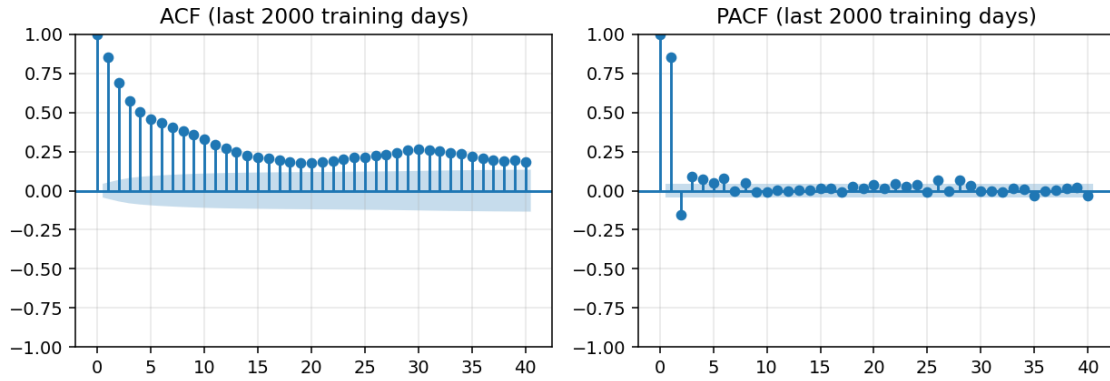


Figure 8: Autocorrelation (ACF) and partial autocorrelation (PACF) of the training wind-speed series (last 2000 training days), used to guide the SARIMA order search.

because the seasonal cycle is strongly bimodal in intensity (monsoon vs. non-monsoon), the annual Weibull fit under-represents the width of the true seasonal spread. A month-by-month or monsoon/non-monsoon Weibull fit would give a more physically faithful resource characterisation for a real siting decision.

L6. No economic analysis. This study addresses the technical wind resource and forecastability, not the economics (capital cost, payback period, grid-connection cost) of an actual small wind turbine installation, which would be required to complete a full feasibility study.

10 Conclusion

This case study demonstrates a complete, reproducible workflow for assessing small-wind-turbine feasibility at a single site using forty-four years of free NASA POWER daily wind data: cleaning and validating the raw record, characterising the long-term resource with a Weibull distribution and a wind-shear height extrapolation, and building and fairly comparing a classical statistical (SARIMA) and a machine-learning (Random Forest) day-ahead forecasting model. For Mumbai, India, the wind resource (mean

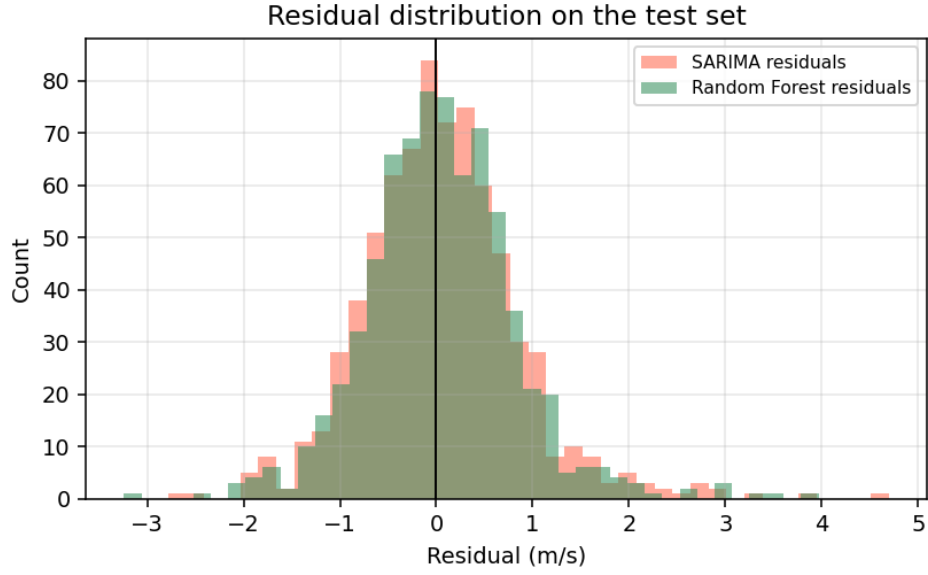


Figure 9: Test-set residual distributions for the SARIMA and Random Forest models.

4.95 m/s at 50 m, Weibull $k = 2.95$, $c = 5.54$ m/s, power density 105 W/m^2) is judged *marginally feasible for small, low-cut-in-speed turbines*, especially during the June–September monsoon season, but *not attractive for utility-scale wind farms*. Of the two forecasting models, Random Forest Regression gave the best one-day-ahead test performance (RMSE = 0.785 m/s, $R^2 = 0.785$), a modest but consistent improvement over SARIMA (RMSE = 0.820 m/s) and a clearer improvement over a naive persistence baseline (RMSE = 0.899 m/s), with the gap between the two models narrow enough that a hybrid ARIMA-plus-Random-Forest design, as demonstrated elsewhere in the literature [5], is a promising direction for future refinement of this specific pipeline. The complete code, data extract, tables, and figures accompanying this report allow the same analysis to be repeated for any other NASA POWER grid cell in the world with only the latitude, longitude, and site name changed.

References

- [1] J. L. Torres, A. Garcia, M. De Blas, and A. De Francisco, “Forecast of hourly average wind speed with ARMA models in Navarre (Spain),” *Solar Energy*, vol. 79, no. 1, pp. 65–77, 2005.
- [2] J. C. Palomares-Salas, J. J. G. De La Rosa, J. G. Ramiro, J. Melgar, A. Agüera, and A. Moreno, “ARIMA vs. neural networks for wind speed forecasting,” in *2009 IEEE International Conference on Computational Intelligence for Measurement Systems and Applications (CIMSAS)*, pp. 129–133, 2009.
- [3] H. Liu, H. Q. Tian, and Y. F. Li, “Comparison of two new ARIMA-ANN and ARIMA-Kalman hybrid methods for wind speed prediction,” *Applied Energy*, vol. 98, pp. 415–424, 2012.
- [4] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] D. Vassallo, R. Krishnamurthy, and H. J. S. Fernando, “Utilizing physics-based input features within a machine learning model to predict wind speed forecasting error,” *Wind Energy Science*, vol. 6, no. 1, pp. 295–309, 2021.
- [6] M. R. Islam, R. Saidur, and N. A. Rahim, “Assessment of wind energy potentiality at Kudat and Labuan, Malaysia using Weibull distribution function,” *Energy*, vol. 36, no. 2, pp. 985–992, 2011.

- [7] F. H. Mahmood, A. K. Resen, and A. B. Khamees, “Wind characteristic analysis based on Weibull distribution of Al-Salman site, Iraq,” *Energy Reports*, vol. 6, pp. 79–87, 2020.
- [8] M. Bortolini, M. Gamberi, A. Graziani, R. Manzini, and F. Pilati, “Performance and viability analysis of small wind turbines in the European Union,” *Renewable Energy*, vol. 62, pp. 629–639, 2014.
- [9] NASA Langley Research Center, “Prediction Of Worldwide Energy Resources (POWER) Project.” <https://power.larc.nasa.gov/>, 2025. Data accessed via the NASA Open Data Portal, <https://data.nasa.gov/dataset/prediction-of-worldwide-energy-resources-power>, daily point time series, Mumbai, India (19.076°N, 72.878°E), 1 Jan 1981–31 Dec 2024.
- [10] J. F. Manwell, J. G. McGowan, and A. L. Rogers, *Wind Energy Explained: Theory, Design and Application*. John Wiley & Sons, 2nd ed., 2010.